

Customer Analytics Report: RFM Segmentation & Cohort Retention Analysis

Executive Summary

This report analyzes customer purchasing behavior using the Brazilian Olist e-commerce dataset. The primary objective is to understand customer value, retention patterns, and revenue concentration using RFM (Recency, Frequency, Monetary) analysis and cohort retention analysis.

The analysis reveals that the business has strong customer acquisition but weak repeat purchase behavior. A small segment of customers contributes a disproportionately large share of revenue. Retention drops sharply after the first month, indicating an opportunity for lifecycle marketing and customer experience improvements.

Business Questions

The analysis is designed to answer the following key business questions:

- Which customers generate the most revenue?
- How frequently do customers return after their first purchase?
- Where does customer retention decline most sharply?
- Which customer segments should be prioritized for marketing efforts?
- What strategic actions can improve long-term customer value?

Dataset Overview

Source: Olist Brazilian E-Commerce Public Dataset.

The analysis combines three primary tables: customers, orders, and payments. After preprocessing and merging, the dataset represents transaction-level purchase behavior for each customer.

Key preprocessing steps included datetime conversion, filtering for delivered orders, handling missing values, and merging datasets on customer and order identifiers.

Methodology

The analysis follows a structured customer analytics workflow:

1. Data Cleaning and Preparation
2. RFM Feature Engineering
3. Customer Segmentation
4. Cohort Retention Analysis

RFM metrics were computed at the customer level. Recency measures days since last purchase, Frequency measures number of unique orders, and Monetary measures total spend. Customers were then segmented using score-based rules to enable business interpretation.

RFM Analysis Insights

Customer purchase behavior is highly skewed. The vast majority of customers make only a single purchase, resulting in an average frequency close to 1. This indicates strong acquisition but weak natural retention.

Revenue is heavily concentrated among higher-spending users. The 'Big Spenders' segment contributes the largest share of total revenue, while a significant portion of revenue also comes from 'Hibernating' customers who are no longer active.

Champions exhibit the highest monetary value and purchase frequency but represent a relatively small portion of the customer base, suggesting an opportunity to expand this segment.

Cohort Retention Analysis

Monthly cohort analysis reveals a sharp drop in retention after the first month. This pattern is typical for transactional e-commerce businesses but highlights a key growth opportunity.

Later-month retention stabilizes at a low level, indicating that customers who do not repeat early are unlikely to return organically. Early lifecycle engagement is therefore critical.

Cohort patterns suggest that improvements in onboarding, post-purchase engagement, and remarketing could materially improve customer lifetime value.

Strategic Recommendations

Based on the analysis, the following actions are recommended:

- Implement loyalty and VIP programs for Champions to protect high-value users.
- Launch targeted win-back campaigns for At Risk and Hibernating customers.
- Provide premium bundles and cross-sell offers to Big Spenders.
- Improve first-month customer experience to reduce early churn.
- Shift part of the marketing budget from pure acquisition toward retention initiatives.

These actions are expected to increase repeat purchase rate, improve customer lifetime value, and stabilize revenue growth.

Future Enhancements

Potential next steps for deeper analysis include:

- Customer Lifetime Value (CLV) modeling
- Churn prediction using machine learning
- Behavioral clustering using unsupervised learning
- Interactive dashboard development
- Cohort revenue and profitability analysis

Customer Churn Analysis Report

Project: Olist E-commerce Customer Analytics

Analyst: Dinesh Khirwar

Churn Definition: Customer inactive for > 90 days

1. Executive Summary

This analysis investigates customer churn behavior in the Olist e-commerce dataset using an inactivity-based churn definition. The primary objective is to quantify churn, identify behavioral risk factors, evaluate segment-level vulnerability, and estimate revenue exposure from customer attrition.

The analysis reveals an **extremely high churn rate (~80%)**, driven largely by a structural one-time buyer pattern. While this is typical for many marketplace businesses, it indicates significant opportunity in early lifecycle engagement and repeat purchase stimulation. High-value customer churn is also present, suggesting potential revenue leakage that can be mitigated through targeted retention strategies.

2. Business Objective

The business aims to answer the following questions:

- What proportion of customers become inactive after purchase?
 - Which behavioral patterns are associated with churn?
 - Which RFM segments are most vulnerable?
 - How severe is the one-time buyer problem?
 - How much high-value revenue is at risk?
 - Where should retention efforts be prioritized?
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3. Churn Definition

Definition Used

A customer is classified as **churned** if:

Days since last purchase > 90 days

Rationale

- The business is non-subscription e-commerce.
- Customers purchase irregularly.
- A 90-day inactivity window is commonly used for marketplace analysis.
- The dataset shows low purchase frequency, making shorter windows too aggressive.

Important Assumption

This churn definition reflects **behavioral inactivity**, not explicit customer cancellation. Therefore, some customers labeled as churned may still return in the future.

4. Overall Churn Rate

Key Result

Overall churn rate: ~80.23%

Interpretation

This indicates:

- Very high customer drop-off after initial purchase
- Weak natural retention loop
- Acquisition-heavy growth pattern

Business Meaning

The platform successfully attracts customers but struggles to convert them into repeat buyers. This pattern is common in marketplace environments but represents a major opportunity for lifecycle optimization.

5. Behavioral Drivers of Churn

5.1 Recency vs Churn

Observation

Customers labeled as churned exhibit significantly higher recency values compared to active customers.

Interpretation

This validates the churn definition and confirms expected behavioral separation. Recency is the strongest immediate indicator of churn risk.

Business Meaning

Customers quickly become inactive after their last purchase, suggesting insufficient post-purchase engagement.

5.2 Frequency vs Churn (Critical Finding)

Observation

- Purchase frequency is heavily concentrated at **1 order per customer**.
- Both churned and active groups show low median frequency.
- The dataset is dominated by one-time buyers.

Key Insight

The business suffers from a severe one-time buyer problem.

Why This Matters

One-time buyers typically:

- have weak brand attachment
- require reactivation nudges
- exhibit high churn probability

Business Risk

Without improving second-purchase conversion, acquisition spend may produce low lifetime value.

5.3 Monetary vs Churn

Observation

High spenders are present in both churned and active groups.

Interpretation

Spending alone does not guarantee retention. Even customers with meaningful purchase value may disengage.

Business Meaning

Retention strategy should not focus only on low spenders — valuable customers are also at risk.

6. Segment-Level Churn Analysis

Key Results (approximate)

- **Hibernating:** ~100% churn

- **At Risk:** ~100% churn
- **New Customers:** ~67% churn
- **Big Spenders:** ~66–67% churn
- **Loyal Customers:** ~67% churn
- **Champions:** ~51–52% churn

Interpretation

Expected High Churn Segments

Hibernating and At Risk segments showing near-total churn is expected because these segments are partially defined by recency deterioration.

Important Risk Signal

The most concerning finding is:

Even Champions and Big Spenders show substantial churn rates.

Business Meaning

High-value users are not fully protected by current customer experience. There is room for:

- loyalty strengthening
- premium retention programs
- lifecycle marketing

7. Frequency Bucket Analysis

Observation

Churn rate is highest among customers with frequency = 1.

Interpretation

This confirms the structural retention weakness:

Customers who fail to make a second purchase rarely return.

Strategic Importance

The **second purchase moment** is the most critical intervention point in the customer lifecycle.

8. Cohort-Based Churn View

Using the cohort retention matrix, churn was computed as:

Churn = 1 – Retention

Key Pattern

- Sharp drop after first purchase month
- Retention stabilizes at a low level
- Later cohorts show similar decay shape

Interpretation

This indicates a classic transactional marketplace pattern:

- strong initial acquisition
- weak habit formation
- limited long-term stickiness

Business Meaning

Early lifecycle engagement is the highest-leverage opportunity.

9. High-Value Customer Churn

Customers above the 75th percentile of monetary value were analyzed for churn risk.

Key Finding

A non-trivial number of high-value customers have churned.

Why This Matters

Not all churn is equal.

Losing high-value customers results in:

- disproportionate revenue loss
- reduced customer lifetime value
- weaker cohort economics

Strategic Insight

Retention efforts should be prioritized using a **value-weighted approach**, not just customer counts.

10. Class Imbalance Assessment

The dataset shows heavy skew toward churned customers (~80%).

Implications

For future churn prediction models:

- Accuracy will be misleading

- Recall and ROC-AUC will be more meaningful
- Class weighting or resampling may be required

Positive Note

This imbalance reflects real marketplace behavior rather than data error.

11. Key Risk Areas Identified

Structural Risks

- Extremely high one-time buyer share
- Weak repeat purchase loop
- High early-lifecycle drop-off

Revenue Risks

- Presence of high-value churners
- Champions not fully retained
- Big Spenders showing notable churn

Behavioral Risks

- Low purchase frequency
 - Rapid inactivity after first order
 - Limited customer habit formation
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12. Strategic Recommendations

Immediate (High Impact)

1. Second-Purchase Acceleration

Focus on converting first-time buyers into repeat customers through:

- post-purchase email flows
 - limited-time repeat incentives
 - personalized recommendations
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2. High-Value Customer Protection

Implement VIP or premium retention programs for:

- Champions
- Big Spenders

- top monetary percentile
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3. Win-Back Campaigns

Target:

- At Risk
- Hibernating

using:

- remarketing
 - email reactivation
 - targeted discounts
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4. Early Lifecycle Optimization

Because churn spikes early:

- improve onboarding
 - enhance first delivery experience
 - optimize first-purchase journey
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5. Retention Measurement Framework

Track:

- repeat purchase rate
 - time to second purchase
 - cohort month-1 retention
 - value-weighted churn
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13. Recommended Next Analytical Steps

To further strengthen customer intelligence:

1. Customer Lifetime Value (CLV) modeling
2. Churn prediction using machine learning
3. Time-to-second-purchase analysis
4. Kaplan–Meier survival analysis
5. Retention uplift experimentation

🚩 14. Conclusion

The analysis reveals that the Olist platform experiences **structurally high churn driven by one-time buyer behavior**, with approximately 80% of customers becoming inactive within the defined window. While this pattern is common in marketplace businesses, it highlights substantial opportunity in early lifecycle engagement and repeat purchase stimulation.

Most critically, churn is not confined to low-value users — meaningful revenue risk exists among higher-value segments. Addressing the second-purchase gap and implementing value-weighted retention strategies represent the highest-impact opportunities for improving long-term customer economics.